US SPACE TRAVEL

A TIMELINE OF MANNED NASA FLIGHTS

On July 21, 2011, the last Space Shuttle flight landed, ending 50 years of manned NASA flights. The US space program began in 1961 and resulted in 166 nearly continuous missions, costing an estimated $7 billion per year. Space travel continues through international partnerships and unmanned missions, but it remains to be seen what is next for American space exploration.

MERCURY

Mercury Redstone 3 (Freedom 7), May 5, 1961
Mercury Redstone 4 (Liberty Bell 7), July 24, 1961
Mercury Atlas , 6 (Friendship 7), February 20, 1962
Mercury Atlas 7 (Aurora 7), May 24, 1962
Mercury Atlas 8 (Sigma 7), October 3, 1962
Mercury Atlas 9 (Faith 7), May 15–16, 1963

GEMINI

Gemini 3, March 23, 1965
Gemini IV, June 3–7, 1965
Gemini V, August 21–29, 1965
Gemini VII, December 4–18, 1965
Gemini VI, December 15–16, 1965
Gemini VIII, March 16, 1966
Gemini IX, June 3–6, 1966
Gemini X, July 18–21, 1966
Gemini XI, September 12–15, 1966
Gemini XII, November 11–15, 1966

APOLLO

Apollo 7, October 11–22, 1968
Apollo 8, December 21–27, 1968
Apollo 9, March 3–13, 1969
Apollo 10, May 18–26, 1969
Apollo 11, July 16–24, 1969
Apollo 12, November 14–24, 1969
Apollo 13, April 11–17, 1970
Apollo 14, January 31–February 9, 1971
Apollo 15, July 26–August 7, 1971
Apollo 16, April 16–27, 1972
Apollo 17, December 7–19, 1972

SKYLAB

Orbited the Earth from 1973 to 1979
Skylab 2, May 25–June 22, 1973
Skylab 3, July28–September 25, 1973
Skylab 4, November 16, 1973–February 8, 1974 (at 84 days, 1 hour, 15 minutes, and 31 seconds, Skylab 4 remains the longest US space flight to date)

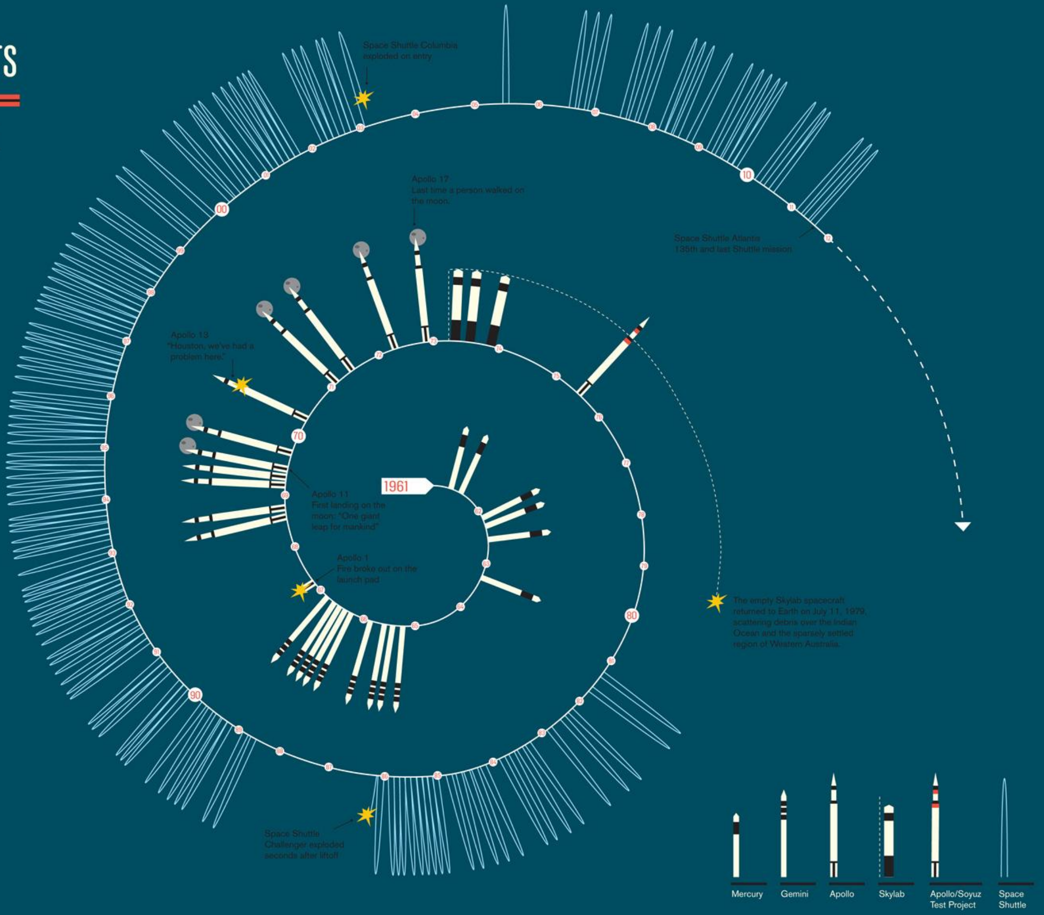
APOLLO/SOYUZ TEST PROJECT

July 15–24, 1975, The first international partnership in space.

SPACE SHUTTLE

Shuttles Columbia, Challenger, Discovery, Atlantis and Endeavour
1981: STS-1, STS-2/1982: STS-3, STS-4, STS-5/1983: STS-6, STS-7, STS-8, STS-9/1984: STS-41B, STS-41C, STS-41D, STS-41G, STS-51A/1985: STS-51C, STS-51D, STS-51B, STS-51G, STS-51F, STS-51I, STS-51J, STS-61A, STS-61B/1986: STS-61C, STS-51L/1988: STS-26, STS-27/1989: STS-29, STS-30, STS-28, STS-34, STS-33/1990: STS-32, STS-36, STS-31, STS-41, STS-38, STS-35/1991: STS-37, STS-39, STS-40, STS-43, STS-48, STS-44/1992: STS-42, STS-45, STS-49, STS-50, STS-46, STS-47, STS-52, STS-53/1993: STS-54, STS-56, STS-55, STS-57, STS-51, STS-58, STS-61/1994: STS-60, STS-62, STS-59, STS-65, STS-64, STS-68, STS-66/1995: STS-63, STS-67, STS-71, STS-70, STS-69, STS-73, STS-74/1996: STS-72, STS-75, STS-76, STS-77, STS-78, STS-79, STS-80/1997: STS-81, STS-82, STS-83, STS-84, STS-94, STS-85, STS-86, STS-87/1998: STS-89, STS-90, STS-91, STS-95, STS-88/1999: STS-96, STS-93, STS-103/2000: STS-99, STS-101, STS-106, STS-92, STS-97/2001: STS-98, STS-102, STS-100, STS-104, STS-105, STS-108/2002: STS-109, STS-110, STS-111, STS-112, STS-113/2003: STS-107/2005: STS-114/2006: STS-121, STS-115, STS-116/2007: STS-117, STS-118, STS-120/2008: STS-122, STS-123, STS-124, STS-126/2009: STS-119, STS-125, STS-127, STS-128, STS-129/2010: STS-130, STS-131, STS-132/2011: STS-133, STS-134, STS-135

SOURCES: www.nasa.gov; www.windows2universe.org/space_missions/manned_table.html; www.rocketmime.com/space/timeline.html; www.freakonomics.com/2008/01/11/is-space-exploration-worth-the-cost-a-freakonomics-quorum/



Space Shuttle Columbia exploded on entry

Apollo 17
Last time a person walked on the moon.

Space Shuttle Atlantis
135th and last Shuttle mission

Apollo 13
"Houston, we've had a problem here."

Apollo 11
First landing on the moon: "One giant leap for mankind"

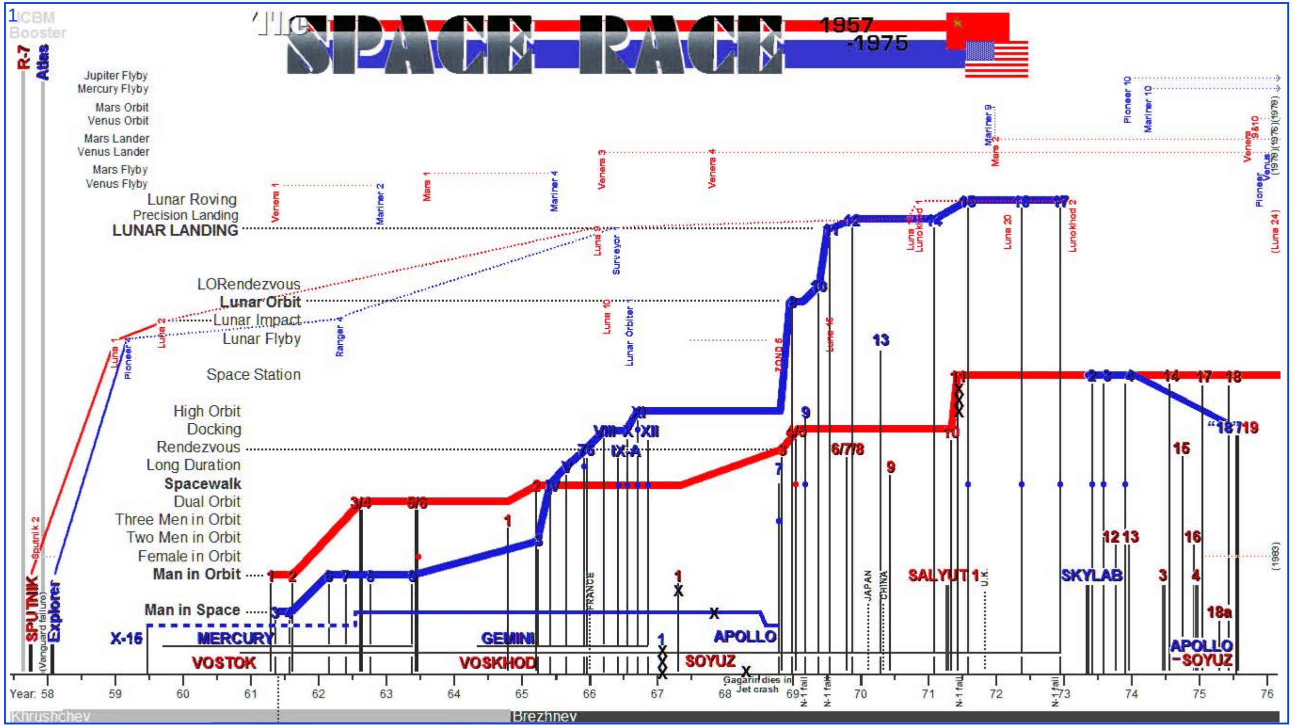
Apollo 1
Fire broke out on the launch pad

The empty Skylab spacecraft returned to Earth on July 11, 1979, scattering debris over the Indian Ocean and the sparsely settled region of Western Australia.

Space Shuttle Challenger exploded seconds after liftoff

MGMT. design

The SPACE RACE 1957-1975



Context:

U.K.'s first hydrogen bomb test
SAC starts 24/7 nuke bomber alert
(continuous until 1991)
USSR unilateral nuke test moratorium
(followed by US and UK)
Thor IRBMs deployed in the UK
Eisenhower creates NASA
Castro overthrows Batista in Cuba
SSBN George Washington launched
Hawaii becomes fiftieth state
France's first nuclear bomb test
Jupiter IRBMs deployed in Italy
U-2 Pilot Gary Powers shot down
Sino-Soviet Split
Nedelin Launch Pad Disaster (kept secret)
Bay of Pigs Fiasco
JFK proposes commitment to land on the Moon
Fallout Shelters/Berlin Wall constructed
Jupiter IRBMs deployed in Turkey
50MT Tsar Bomba detonated
Telstar relays first live TV by satellite
CUBAN MISSILE CRISIS
Nuclear Test Ban signed by US, USSR & UK
JFK proposes cooperation on Apollo (UN speech)
JFK Assassinated

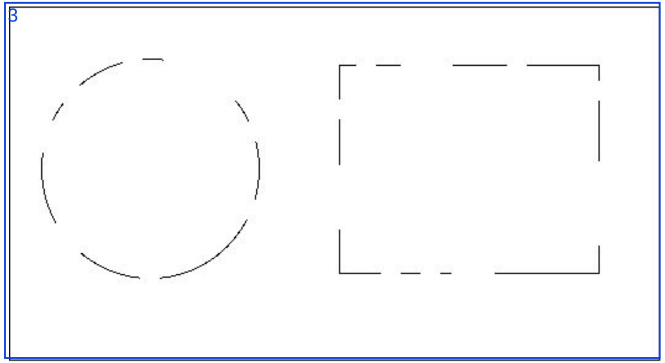
U. S. in Vietnam War

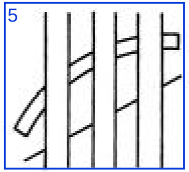
U.S. Civil Rights Act
China's first nuclear bomb test
(joins U.S., U.S.S.R., U.K. & France)
Winston Churchill dies (stroke)
U.S. invades Dominican Republic
Malcom X Assassinated
Sergei Korolev dies after surgery
Color TV becomes popular
First Super Bowl
Stalin's daughter defects to the U.S.
Interracial marriage legalized for U.S.
Czech reform (stopped by Warsaw Pact invasion & occupation thru 1990)
MLK Assassinated
RFK Assassinated
Trousers accepted for Women's everyday wear
Eisenhower dies (heart failure)
Chappaquiddick
R-36 FOBS deployed
Mylai Massacre
Nuclear Non-Proliferation Treaty
3 Jets bound for NY Hijacked
4 Kent State war protesters killed
Intel 4004 first microprocessor
Khruschev dies (heart failure)
HP-35, first scientific pocket calculator
Nixon visits China and Russia
SALT I marks beginning of **DETENTE**
Fischer beats Spassky in Chess Match
Munich Olympic Terrorism
Oil Crisis
LBJ dies (heart failure)
Tu-144 crashes at Paris Air Show
Billie Jean King beats Bobbie Riggs
Solzhenitsyn exiled
India's first nuclear bomb test
Nixon resigns after Watergate Scandal
Ella Grasso elected as female governor
Sony markets Betamax
Atari markets Pong home console

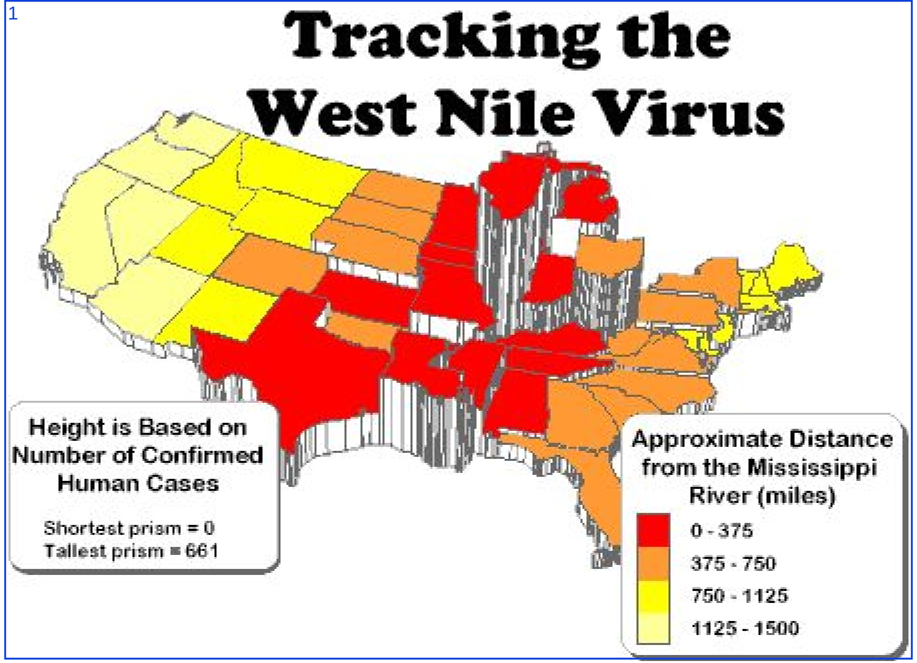
Gestalt Principles of Visual Perception

Closure. We tend to perceive objects such as shapes, letters, pictures, etc., as being whole when they are not complete. Specifically, when parts of a whole picture are missing, our perception fills in the visual gap.

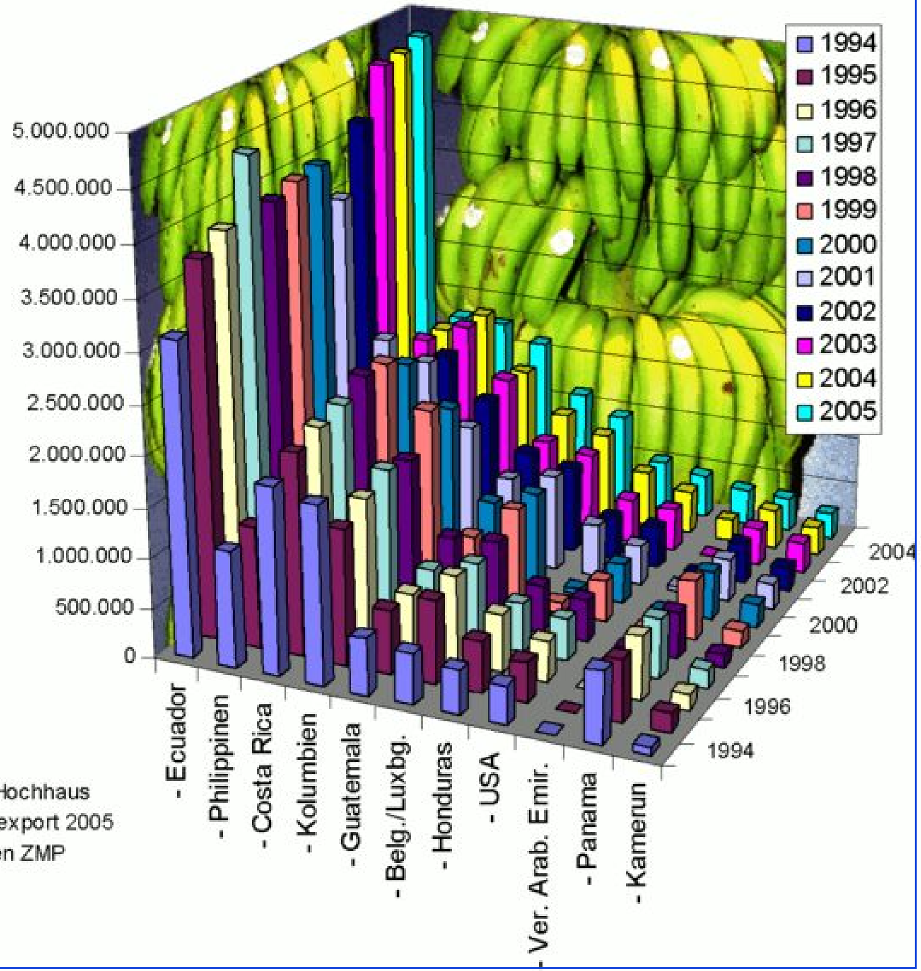
Continuity. Elements of objects tend to be grouped together, and therefore integrated into perceptual wholes if they are aligned within an object. In cases where there is an intersection between objects, individuals tend to perceive the two objects as two single uninterrupted entities.






Tracking the West Nile Virus
Height is Based on Number of Confirmed Human Cases
Shortest prism = 0
Tallest prism = 661
Approximate Distance from the Mississippi River (miles)
0 - 375
375 - 750
750 - 1125
1125 - 1500

Export von Bananen in Tonnen von 1994-2005



Dr. Hochhaus
Banexport 2005
Daten ZMP